END SEMESTER EXAMINATION– 2025

Name of the Program: M.Tech
Name of the Course: Computer Vision and its Applications
Time: 3 Hour

Semester: 2nd
Course Code: MCS-241
Maximum Marks: 100

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 20 marks

Q1 (20 marks)

(a) Define low-level, mid-level, and high-level vision tasks. Classify the following applications accordingly: edge detection, object recognition, and scene understanding. Justify each classification. **CO1**

(b) Compare and contrast Computer Vision with Image Processing and Computer Graphics. Support your answer with relevant examples from biometrics, medical imaging, and augmented reality. **CO1**

(c) Discuss the challenges of building real-time computer vision systems. Choose two diverse applications (e.g., content-based image retrieval and object tracking) and explain their specific challenges and possible solutions. **CO2**

Q2 (20 marks)

(a) Explain the significance of bit-plane slicing in image enhancement. Illustrate how different bit-planes contribute to the overall structure and perception of a grayscale image using a sample 8-bit image. **CO2**

(b) Discuss gamma correction and its role in brightness normalization. Apply different gamma values ($\gamma < 1$, $\gamma = 1$, $\gamma > 1$) on a sample intensity range and interpret the results graphically. **CO2**

(c) Design spatial filters for image smoothing and sharpening. Apply a box filter and a Laplacian kernel on a 5×5 image matrix, and compare the results in terms of edge preservation and noise suppression. **CO3**

Q3 (20 marks)

(a) Explain the role of image filtering in noise removal and feature enhancement. Differentiate between linear filters and non-linear filters, including their effects on edge preservation. **CO3**

(b) Design a morphological filtering technique for removing salt-and-pepper noise from binary images. Explain the sequence of operations (e.g., opening followed by closing) on a sample 5×5 binary matrix. **CO4**

(c) What are image thresholding techniques? Compare global thresholding, Otsu's method, and adaptive thresholding. Use a sample histogram to demonstrate each. **CO5**

Q4 (20 marks)

(a) What is background subtraction in video processing, and how does it aid in detecting moving objects? Additionally, explain the concept of frame differencing with the help of a simple example using three consecutive video frames to demonstrate how motion is detected. **CO4**

(b) What is object recognition in computer vision? Explain its purpose and importance in real-world applications. Provide one simple example of object recognition using a deep learning method (such as identifying letters, human faces, or traffic signs). **CO5**

(c) What is the difference between Augmented Reality (AR) and Mixed Reality (MR)? Explain the key concepts, features, and use cases of both AR and MR. Support your answer with real-world examples (e.g., AR in education, MR in surgery simulation). **CO6**

Q5 (20 marks)

(a) Design a CNN architecture for image classification. Explain the function of convolutional layers, activation functions, pooling layers, and fully connected layers. Use a 32×32 input image as a reference. **CO5**

(b) Describe a real-time activity recognition system using computer vision. Outline the pipeline from video acquisition to action classification such as video anomaly detection using deep learning. **CO6**

(c) A research team is developing a computer vision-based system to assist radiologists in clinical decision-making by fusing multimodal medical images, such as MRI (anatomical) and PET (functional). The goal is to preserve critical structural and functional details while enhancing diagnostic visibility. To achieve this, the team plans to use guided filtering as a core fusion technique, supplemented by both traditional image processing and deep learning strategies for enhancement and evaluation. Based on this case study, design and discuss complete computer vision architecture for multimodal medical image fusion using guided filtering. Additionally, analyze the critical factors that influence the performance of medical image fusion systems, particularly in the context of clinical image analysis. **CO6**